

MOHAMMED FERWANA

Backend Engineer · System Architecture

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PROFESSIONAL SUMMARY

Methodical and reliability-driven Backend Engineer with demonstrated ownership in architecting high-integrity server-side systems, finite state machine (FSM) workflows, and scalable RESTful APIs. Experience leading backend engineering delivery, designing relational and document data schemas (PostgreSQL & MongoDB), enforcing strict role-based access control (RBAC), and authoring comprehensive automated integration test suites. Solid foundation in Computer Systems Engineering principles, algorithm design, and distributed patterns.

TECHNICAL SKILLS & COMPETENCIES

Backend Runtimes: Node.js, Express.js, RESTful API Design, Microservices Concepts, Event-Driven Patterns

Databases & Storage: MongoDB, PostgreSQL, Mongoose, Sequelize, Redis (Caching / Pub-Sub), Compound Indexing

Architecture & Security: Finite State Machines (FSM), Multi-Tenant Isolation, JWT, RBAC, Data Validation, Rate Limiting

Testing & Quality: Jest, Supertest, In-Memory DB Replicas, Integration Testing, Continuous Integration (CI)

DevOps & Tools: Docker, Gitflow, GitHub Actions, Postman, Linux / Bash, Clean Architecture

PROFESSIONAL EXPERIENCE

Backend Developer & Team Leader — TAQAT (TeamLine Platform) 2024 · Remote

- Architecture & Delivery:** Spearheaded backend system architecture for TeamLine, a collaborative project management system, ensuring predictable RESTful API contracts across teams.
- Security & Access Control:** Designed enterprise-grade authentication with JWT and granular workspace-scoped Role-Based Access Control (RBAC).
- Engineering Leadership:** Conducted sprint planning, technical task estimation, and peer code reviews, accelerating engineering velocity and code consistency.
- System Reliability:** Established centralized error handling pipelines, request sanitization, and database indexing that eliminated duplicate records.

Market Ready Developer Program — Gaza Sky Geeks (GSG) 2023 – 2024 · Gaza, Palestine

- Completed rigorous professional immersion centered on full-stack architecture, clean code principles, and PostgreSQL relational database design.
- Architected modular services adhering to separation of concerns, test-driven validation, and professional Gitflow branching workflows.
- Collaborated in pair programming, architectural reviews, and agile sprint cadence.

FEATURED ENGINEERING SYSTEMS & PROJECTS

InsurFlow — B2B Motor Insurance Claim Lifecycle Platform Backend Owner · 2024 – Present

- Finite State Machine:** Designed and implemented a deterministic 9-state claim lifecycle state machine preventing invalid status mutations with explicit 409 conflict handling.
- Multi-Tenant Compound Indexing:** Structured tenant-scoped data isolation in MongoDB using compound B-tree indexes with organizationId leading keys to guarantee zero data leakage.
- Atomic Concurrency:** Engineered sequential claim numbering (CLM-ORG-0001) via atomic counter operations, verified collision-free under 50-request concurrent bursts.
- Automated Integration Testing:** Authored 40+ automated integration tests across comprehensive test suites with Jest and MongoMemoryServer, validating business invariants.

TeamLine — Real-Time Collaborative Workspace Engine Lead Backend Engineer

- Built modular REST API services for workspace membership, task assignment, and activity tracking.
- Enforced schema validation and indexing that sustained rapid query response times under high-volume queries.

EDUCATION

- Core Coursework: Operating Systems, Data Structures & Algorithms, Database Management Systems, Computer Networks, Software Engineering, Discrete Mathematics.